

Accelerating Reinforcement-Learning-Based Subgrid-Scale Stress Model Development with Agentic AI

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Turbulence modeling strategies are of vital importance in the field of computational fluid dynamics (CFD) because they allow the practitioner to perform simulations of a turbulent flow at reduced cost compared to solving the full-fidelity Navier-Stokes equations. The problem is that coarse-graining the Navier-Stokes equations to achieve this reduced cost results in a closure problem. This means that the coarse-grained system has more unknowns than equations and so post-hoc closure relations must be designed to model the unresolved physics.

In this project in particular, we are interested in the so-called Large-Eddy Simulation (LES) methodology in which only the large scale turbulent structures are directly resolved and the small scales are left unresolved (their effect must be accounted for by the closure model). Currently, the industry standard is to use these subgrid-scale (SGS) stress models in conjunction with special near-wall models (which are specifically designed to model the behavior of turbulent flows near a solid wall where grid resolution requirements can easily become prohibitive). Importantly, the models commonly used in industry were developed decades ago, and do not directly account for the complex flow physics that are commonly encountered in engineering designs. Therefore, the purpose of this research is to improve the accuracy of SGS models through the use of modern reinforcement learning (RL) algorithms.

The reason for using RL in this project is because it allows the model to be trained within the context of an actual LES environment. Crucially, by incorporating a pre-trained machine-learning-based wall model that has been previously developed by members of our group within the RL environment, the obtained SGS model will be automatically consistent with that wall model. Therefore we expect to realize the benefits of improved wall modeling and SGS modeling simultaneously due to the use of this training methodology.

What I Expect the Agent to Do: I have already been working on this project and have been using coding agents to implement the integration of an LES solver with an existing RL library. Therefore at a baseline I expect the model to continue to help me build out and verify the implementation of the infrastructure needed to perform the RL training.

At the same time, I believe there are several ways that I could use agentic AI to further streamline the development of the actual RL approach I am using to train these models. For example, I would like to be able to largely automate the running of different numerical experiments with specially designed agents. Ideally, I (as the human user) would like to remain in the development loop as an orchestrator, but I believe that the AI is likely much better (and certainly faster) than I am at configuring experiments and distilling results. Therefore, my goal is to be able to make significant progress on my project through the effective use of these agentic tools.